Pedigrees

A pedigree table (family tree) is often used in genetics to show how characteristics are passed from one generation to the next. A pedigree is usually set out with the earliest known ancestors at the top of the table and succeeding generations in line of descent including direct and collateral descendents. A pedigree is especially useful in tracing the transfer of sex-linked characteristics. There are many conventions used in drawing up these tables.

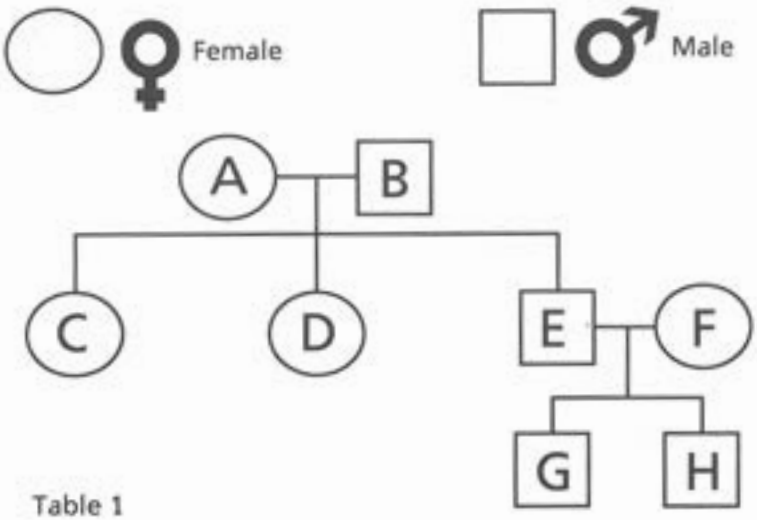


Table 1

Table 1 shows a particular human family. Female A marries male B and has three offspring, C (female), D (female) and E (male).

E marries female F and this couple have two male offspring, G and H.

1 Use a similar system to draw up a pedigree for the following family:

A male and female married and had two offspring, one boy and one girl. The male never married and had no offspring. The female married and had three children, two boys and one girl. All of these children subsequently married. The first male of this generation had two female offspring, the second male had no offspring and the female had one boy and one girl.

2 Explain the difference between *direct* and *collateral* descendents.

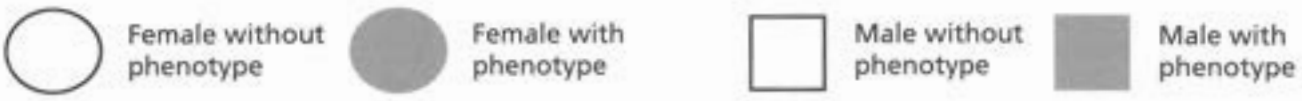
Phenotypes are the physical characteristics resulting from inherited genotypes. Pedigrees can be used to illustrate how physical characteristics occur in generations of a family.

3 What is the name of the technique used to predict possible genotypes and phenotypes of offspring?

4 Explain what is meant by F_1, F_2 and F_3 when referring to generations.

Table 2 shows the occurrence of lobed ears in a family.

5 Describe what is shown by this table.



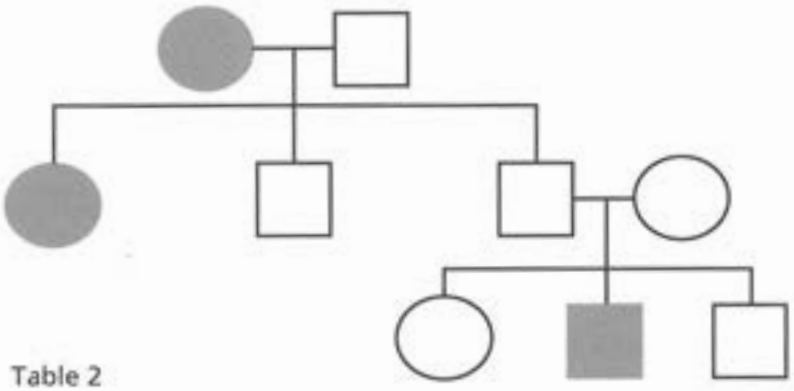


Table 2

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Sex-linkage

Humans characteristically have two sex chromosomes. Male are XY and females are XX. The Y chromosome is smaller, has a different shape to the X chromosome and carries many fewer genes. The Y gene lacks partners for many of its genes. Genes that are unpaired (ie exist only on the X chromosome) are sometimes referred to as *sex-linked genes*. Many inherited defects are due to recessive genes.

A 'carrier' is a person who is carrying a recessive gene for a disease but is not affected by that condition. If the dominant gene is 'normal', it overcomes the recessive gene. Because the Y chromosome has 'missing' genes, sometimes there may not be a dominant gene present and the characteristic carried by the recessive gene may show up. It is possible for a male to suffer from a sex-linked disease by inheriting a single recessive gene (on the X chromosome) for that condition from his mother.

6 Explain the difference between *dominant* and *recessive* genes.

7 Why is it less likely for a female to suffer from a disease due to faults in the X chromosome?

8 A carrier for a disease is likely to be heterozygous for that disease. Define *heterozygous*.

Sex and gender

The terms *sex* and *gender* are often confused. Sex refers to how we differentiate organisms by the part they play in the reproductive process. We usually refer to the organism that donates genetic information as male and the organism that receives genetic information as female. Even bacteria can be divided into male and female on the basis of how they swap chromosomal material. More complex organisms have *gonads* – reproductive organs that differ between males and females. *Hermaphrodites* are organisms that possess both male and female reproductive systems. In mammals, true hermaphrodites are rare. Generally, one of the reproductive systems is vestigial. Most hermaphrodites are sterile, however, some sperm and eggs can be produced (far less sperm). Many of the gametes produced are unviable though births have occurred. More than half of human hermaphrodites are raised as males. Some organisms are also capable of *parthenogenesis* (*Gk* virgin birth) where unfertilised ova can grow into adults genetically identical to the sole parent.

Gender is a society-based classification of males and females. This could be the expectation that (in some cultures) women wear skirts and have long hair, raise children etc and men wear trousers, have short hair and go out to work. Different societies will have different ideas (often stereotypes) as to what are 'male' and 'female' roles.

What do the following terms define?



9 *An organ that makes gametes (sex cells).*

10 *An organism that has both ovaries and testes.*

11 *The development of an organisms from an unfertilised gamete.*

12 Give two characteristics that differentiate males and females by sex.

13 Give two characteristics that differentiate males and females by gender.

14 Define *vestigial* and *unviable*.

15 Suggest why many hermaphrodites are raised as males.

Some physically healthy males and females experience other gender identity issues. These can arise when an individual feels mentally like the opposite sex, or are confused as to which sex they were meant to be or which gender they feel themselves sexually attracted to.

Give definitions for the following gender recognition issues:

16 Homosexual

17 Transvestite

18 Transsexual

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